# HDOC Day-48

**C Programming Activity**

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|---|---|---|---|

## Problem Statement

Write a C program that reads the size and elements of a variable-length integer array **arr[]**. Pass the array to a function named **pivot_index()** using its address and pointers. For each index, calculate the sum of elements strictly to its left and strictly to its right. Display the leftmost index where both sums are equal; otherwise display **-1**.

## Algorithm

1. Read the size n and the n array elements.

2. Pass the array address and n to pivot_index().

3. For each index i, initialize left_sum and right_sum to 0.

4. Add all elements before i to left_sum.

5. Add all elements after i to right_sum.

6. If left_sum equals right_sum, print i and return immediately so the leftmost pivot is displayed.

7. If no index satisfies the condition, print -1.

## C Program

```
#include <stdio.h>

void pivot_index(int *arr, int n)
{
    int i, j;
    int left_sum, right_sum;

    for (i = 0; i < n; i++)
    {
        left_sum = 0;
        right_sum = 0;

        for (j = 0; j < i; j++)
            left_sum = left_sum + *(arr + j);

        for (j = i + 1; j < n; j++)
            right_sum = right_sum + *(arr + j);

        if (left_sum == right_sum)
        {
            printf("Pivot index: %d\n", i);
            return;
        }
    }
    printf("-1\n");
}

int main()
{
    int n, i;
    printf("Enter the size of array: ");
    scanf("%d", &n);
    int arr[n];

    printf("Enter the array elements:\n");
    for (i = 0; i < n; i++)
```

```
        scanf("%d", &arr[i]);

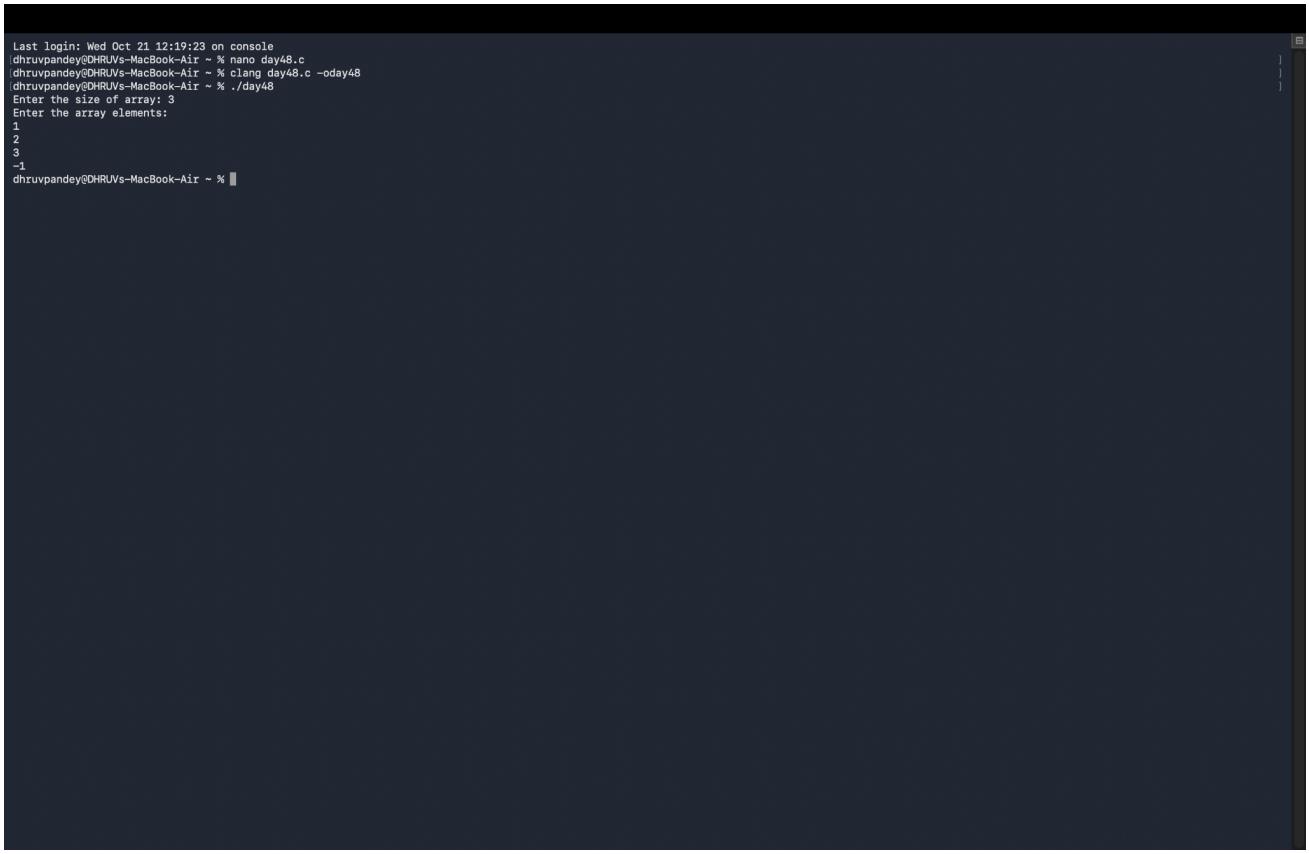
    pivot_index(arr, n);
    return 0;
}
```

## Test Cases

| Test Case | Array | Expected Output |
|---|---|---|
| 1 | 1 7 3 6 5 6 | Pivot index: 3 |
| 2 | 1 2 3 | -1 |
| 3 | 2 1 -1 | Pivot index: 0 |

## Execution

Program compiled and executed in the terminal. The entered array was **1 2 3**, and the program correctly displayed **-1**, since no pivot index exists.




**Result:** The program successfully finds the leftmost pivot index using pointer-based array passing and displays -1 when no pivot index exists.